Choosing what options to provide survey respondents seems…tricky. What do you think should be the "best practices" here?

Gina: "In STAT 421 we were taught that survey options need to be exhaustive, meaning that any option a respondent would want to pick needs to be on the survey. This is clearly bad survey design and needs to be fixed but it is so common."

Charlotte: "I think of the census definitions of race and how exclusionary they are. "White" includes anyone with backgrounds from Europe or Middle East or North Africa. That covers a vast amount of land, cultures, and identities that are not well represented under a single term. The census data informs a lot of policy and structural changes in the US. By reducing people to one of five racial groups, people with unique experiences (ie Middle East, multi racial, other groups not included in the main groups) are left out of systemic changes."

Evie: "So I guess the bottom line is how to not eliminate classification, but how to design it in ways that acknowledge diversity without reproducing the same patterns of exclusion and power imbalance."

Evie: Too many options may overwhelm respondents, but too few options may give us too little information. There needs to be consideration for reducing survey fatigue.

Antonio: Sorting text into categories for more open-ended questions is similar to machine learning. It gives people the illusion of more choices. There is still data being stored in the background. We have the flexibility to recategorize shifting groups.

Justin: Map different spellings of gender (e.g., different capitalization) to a default one. Aggregate to one category (machine learning).

Charlotte: In the reading, it was stated that gender should only be collected for healthcare or other important industries. There are a lot of companies that collect pointless data. We should only collect these types of data when necessary.

Evie: In marketing, we tend to ask survey demographics for customer segmentation, which could be helpful for the company.

Dr. T: Categorical variables are so powerful – they make your analysis super easy. They allow you to talk about difficult to measure things to put into a similar bins, but we may lose information. We must figure out how much information we are willing to lose.

Khoa: to be subjective, there could be an objective scale. The hedonic scale (1 – 9) can be an optimal scale to gauge tests. There can be an optimal number?

Dr. T: Anytime you are compressing something with a range of spectrum, there is a variety of ways that people can take. The validity of that scale – when they measure a scale – a 3 measures that specific type of experience. If we are going to represent race in a certain number of categories, then, we put a category of 'blank,' then we know for sure that thing is measuring the thing we are interested in . A good example would be providing binary gender – someone who is nonbinrary may refuse to take the survey or just pick a category – are they more femme or masc? Where do they fit on that binary. There are men and women and nonbinary people that sorted themselves in that category. Where do trans people put themselves? Where they currently are (transitioned), or what they were when they were born. Our gender bin – that we have made so corse – has extra pieces. The data we want and the options that we give people feels very tricky.

Justin: Would a text response give them more options?

Dr. T: That is the best/most inclusive option. Not sure how complicated it would be to clean messy text data. There could be the text box you could fill in, which looks like its inclusive, but the company BTS could parse it into binary data (whites or nonwhites, males or females). What options seem the most genuine? What are we going to do with this information?

In statistics classes we are typically given data which we took no part in collecting with which we are tasked with analyzing. How can we and the "end users" of the data obtain the context that is necessary to understand any biases that might exist?

Dr. T: When I was thinking of this from a data-user perspective, if we are working for a company they are going to give us the data. We don't often get the experience of collecting our own data.

Erica: "In quantitative research methods as well as my other statistics classes, I rarely got to know the context and story behind the data, as **I only saw the numbers** that were made available through a closed-ended survey."

Charlotte: "This example from the *Telegraph* illustrates how **you can still work around biases** even with binary data."

Justin: You can ask how the survey was distributed.

Dr. T: Who took the survey, but not *why* they chose the options for the survey.

Charlotte: In the telegraph, people doing the design did not reinforce gender stereotypes with pink blue but instead purple green so readers can change their perspectives a bit.

Dr. T: Trying to figure out what agency they have, data was collected in a way that I wouldn't collect them, but what are ways that could further not do harm (visualizations – pink for girls and blue for boys). I find myself worrying, given the current job crisis, when we are in entry level positions, we can't tell the boss we don't want to do something. We can ask questions in a softer way, and take something that is not a great situation and repurpose it to make good out of it, which makes it genuine to you.

Suppose you are working for a company that is interested in collecting data to understand employee's experiences working for the company. The survey they sent you asks each employee for their gender, with options "male" and

"female." What questions would you ask the company? What suggestions would you provide?

Khoa: "I love this point raised about not collecting data leads to invisibility of some communities, where visibility can drives a lot of helpful insights to fight oppression. However, excessive visibility in many cases can also leads to oppression and mass surveillance. The main point being I think collecting data should start with **understanding the purpose and goals** from there we can decide whether it justify overall good, or harm like how experimentation on humans must have board approval."

Charlotte: State your question with intent and purpose to let them know the issue.

Antonio: Asking for intentions behind specific questions or options in the dataset offers a good insight behind the data that was collected and what was trying to achieve.

Khoa: If you have 2 – 3 clients in the same industry, you can't use the same data for the same client. You got to understand your responsibility that you cannot mix data. Having guardrails or sensitive data that is really important. We can establish a framework of what data is okay to use.

Charlotte: What is your experience as a nonbinary person in statistics?

Dr. T: Gender is very fluid. Things have been evolving. It hasn't always the way it feels now. When I was in grad school, my first experiences having this conversations was with someone in academia that had really good intentions. They collected data on gender and only gave binary options. The setting – Montana- has the highest suicide rates in the country – very rural – the research they were doing was like mobile mental health assistance and put on a play and normalized these sorts of feelings and talking about them. They had a pre-post survey and came to the center to analyze the data – there is a segment of the population we are missing – trans and nonbinary people who have the highest rate of suicide – no way to talk about them when we don't have their data. I have been trying to evolve in the racial space – I have always been the default in the space as a White person. The hardest is being misgendered; it happens all the time. It's been a lot of trying to teach people that they can't assume based on appearance. We have a big cultural difference between our generations (us vs our parents generations). Things are evolving and it is a lot of assuming best attentions. When you are hurt, its hard, but no one intends to hurt people. When collecting binary data, its not like they were trying to hurt or exclude trans/nonbinary people. People want to do the most good (maybe not corporations), but being okay with people not being perfect. People can occasionally slip up and use the wrong pronouns, and that doesn't mean they are bad. I mess up calling students the wrong name. We all have times where we are not perfect. It's hard but I think people are good. I try to assume the best.

Erica: To provide a more soft-blow answer, provide more options and give examples to make the conversation less tricky.

Mira: I wonder if you could approach the conversation as , "Do you think this would be helpful, it was used in this similar study?"

Charlotte: The census especially, is a good standard to point to, but only has five categories. I had to deal with select all that apply, in STAT 421, and it was difficult to wrangle and summarize. Is there an easier way to approach this or summarize this rather than selecting how many as possible.

Dr. T: I don't think there is a best practices for it (racial categories for mixed race people). Our data cleaning will be a little bit messy.